4.3 - Optimization

Goals: Optimize the value of a quantity in a word problem
Before Class Videos: 4.3.1, 4.3.2
Note: This file is labeled Oct 30th, but we will likely only just start this worksheet on Friday, and will do most of it Nov 3rd

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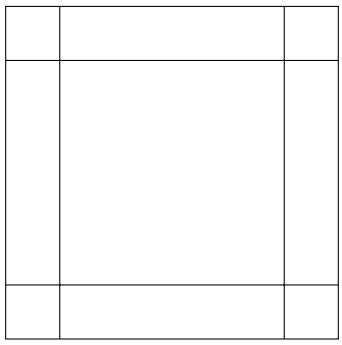
Example 1: A farmer wants to build a rectangular enclosure along a river and has 500 m of fencing for the other three sides. What is the maximum area of the enclosure?

Optimization Steps

1. Draw a diagram. Label relevant parts with variable names.
2. Create an optimization equation involving the variable you want to optimize.
3. Create constraint equations between variables.
4. Use constraint equations to write the optimization equation in terms of one variable.
5. Find (and verify!) extrema.
6. Answer problem.

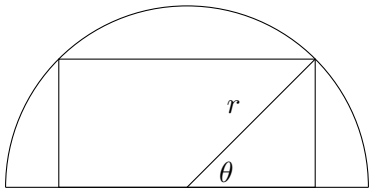
Example 2: An open-top rectangular container has volume 10 m^3. Its length is twice its width. The base material costs \$20/m^2 and the sides cost \$9/m^2. What dimensions minimize the cost?

Example 3: An open-top box is made from a square of side length 8 by cutting squares of size x from the corners.



Example 4: Find the point on $y = \sqrt{x}$ closest to $(3, 0)$.

Example 5: A rectangle is inscribed in a semicircle of radius r.



Example 6: A wire of length L is cut into two at a point x. The left portion is formed into a square. The right portion is formed into a circle. Compute the value of x that maximizes the enclosed area.

Example 7: You went for a swim and need to return to the campground. It is 200 m to swim directly to shore, and then the camp is 1000 m along the shoreline. You can swim at 3 km/h and run at 10 km/h. To what point along the shoreline should you aim so that you minimize the total time to get back to camp?

Suggested after class problems from APEX: Section 4.3: 4, 7, 13, 15